RetailIQ – Full-Cycle E-Commerce Customer & Sales Analytics

DATA SCIENCE (24AD3107) Project Review-1

17/9/2026

Under the Guidance of

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Outline of Project Work

- Focus
- Introduction
- Literature Survey
- Challenges
- Research Objectives
- Conclusion & Future Work
- References
- Publications Details

Problem Statement

- E-commerce businesses generate large amounts of transaction data.
- Raw data is difficult to understand and analyze directly.
- Businesses need insights into sales and customer behavior.
- Product performance and country-wise sales need to be analyzed.
- Businesses also need to understand future revenue trends.
- An integrated analytics solution is needed for better business decision-making.

Objectives

- To clean and preprocess e-commerce transaction data.
- To analyze sales, customer, product, and country-wise performance.
- To perform RFM analysis for understanding customer behavior.
- To segment customers into meaningful groups.
- To forecast future revenue using historical sales patterns.
- To develop an interactive Streamlit dashboard for visualizing insights.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation / Future Work
Fauzan & Alfian [1]	*Customer Segmentation Based on RFM and K-Means*	• RFM Analysis • K-Means Clustering • E-commerce Dataset	• RFM helps identify different customer groups. • Segmentation supports targeted marketing strategies.	• Uses limited customer behavior features. • Additional customer attributes can be considered.
Suresh et al. [2]	*Meta-LLSTM: Meta-Learning Enhanced Learnable LSTM for Retail Sales Forecasting*	• LSTM-based forecasting • Meta-learning • Retail sales data	• ML can capture sales patterns. • Forecasting supports sales and business planning.	• Deep learning models can be complex. • Model interpretability can be improved.
Prakoso & Nuryuliani [3]	*Application Development of E-Commerce Business Intelligence Dashboard*	• Business Intelligence • Data Processing • Interactive Dashboard	• Dashboards convert raw data into useful information. • BI supports business decision-making.	• Mainly focuses on visualization. • Predictive analytics can be added in future.

Problem Gap Identification

Existing Approach	Identified Gap	RetailIQ Solution
Sales data is analyzed separately.	Lack of an integrated analysis of sales, products, customers, and countries.	Provides a combined e-commerce analytics solution.
RFM is used for customer segmentation.	Customer segmentation is often treated separately from sales analysis and forecasting.	Integrates RFM-based segmentation with overall business analytics.
Sales forecasting is performed using historical data.	Forecasting results are not always presented together with customer and sales insights.	Combines revenue forecasting with other analytics in one dashboard.
Business dashboards mainly focus on visualization.	Limited integration of customer segmentation and predictive insights.	Provides an interactive Streamlit dashboard with segmentation and forecasting.
Data analysis may require separate tools or processes.	Difficult for users to view all important insights in one place.	Presents KPIs, trends, customer segments, product insights, and forecasts in one platform.

Innovation, Creativity, and Novelty

Aspect	RetailIQ – Innovation / Novelty
Integrated Analytics	Combines sales, customer, product, and country-wise analysis in one solution.
Customer Segmentation	Uses RFM analysis to group customers into meaningful segments such as Champions, Loyal, At Risk, and Lost.
Revenue Forecasting	Uses historical monthly revenue patterns to estimate future revenue trends.
Interactive Dashboard	Presents KPIs, analysis, customer segments, and forecasts through an interactive Streamlit dashboard.
End-to-End Approach	Provides a complete flow from data cleaning → analysis → segmentation → forecasting → visualization.
Business Decision Support	Converts raw e-commerce transaction data into actionable insights for better decision-making.

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RetailIQ – Feasibility and Project Plan

1 WEEK

1

Phase 1: Planning & Data Collection

- Problem definition
- Dataset collection
- Requirement analysis
- Technology selection



1–2 WEEKS

2

Phase 2: Data Cleaning & Preprocessing

- Missing value handling
- Duplicate removal
- Return/invalid transaction handling
- Date conversion



1–2 WEEKS

3

Phase 3: Data Analysis & Customer Segmentation

- Sales & product analysis
- Customer analysis
- RFM analysis
- Customer segmentation



1–2 WEEKS

4

Phase 4: Revenue Forecasting & Dashboard

- Monthly revenue analysis
- Revenue forecasting
- Streamlit dashboard development
- KPI and visualization integration



1 WEEK

5

Phase 5: Testing & Finalization

- Dashboard testing
- Result validation
- Documentation
- Final review & presentation

